

Turning Ocean Overgrowth into Regenerative Value

ORIGIN OCEAN

SPHERE — REGENERATIVE

Origin by Ocean is a Finnish biotech company turning invasive marine algae into valuable bio-based ingredients. Its patented biorefining technology creates alternatives to chemicals traditionally made from fossil resources, while helping address the environmental damage caused by algal overgrowth.

The company is raising funding for building its first-of-a-kind (FOAK) biorefinery in Kokkola, Finland.

€17M
RAISED TO DATE

€65M
PLANNED INVESTMENT TO SCALE

€51M
TARGET ANNUAL REVENUE

WHY IS IT REGENERATIVE

HARVEST - Coastal entrepreneurs collect invasive algae, taking nitrogen and phosphorus out of the sea before it rots.

REFINE -Closed-loop biorefinery with non-toxic process chemicals and clean energy.

REPLACE -100% bio-based, biodegradable ingredients take oil-based chemicals out of everyday goods.

RETURN -Residue becomes feed, concrete fillers, peat alternatives and biochar that stores carbon.

PROVEN POTENTIAL

from BELIEF *to* BANKABILITY

An Ocean Problem. A New Industrial Resource.

Invasive Sargassum seaweed is accumulating in large volumes, damaging marine ecosystems and affecting coastal communities. At the same time, industries such as cosmetics, food and personal care need alternatives to fossil-based ingredients.

40M t
AVAILABLE SARGASSUM

€22M
OFFTAKE TO FUTURE PRODUCTION

€440M
CUSTOMER PIPELINE

Build the first industrial biorefinery in Kokkola and demonstrate that a regenerative algae-based process can deliver reliable production and attractive economics at scale.

ORIGIN BY OCEANS

LIFE CYCLE IMPACT

1. BIOMASS AS FEEDSTOCK

Harvesting invasive seaweed species and Cultivating native seaweed species.

We remove nitrogen and phosphorus from the oceans by using invasive algae before it decomposes and emit greenhouse gases.

2. OUR BIOREFINERY

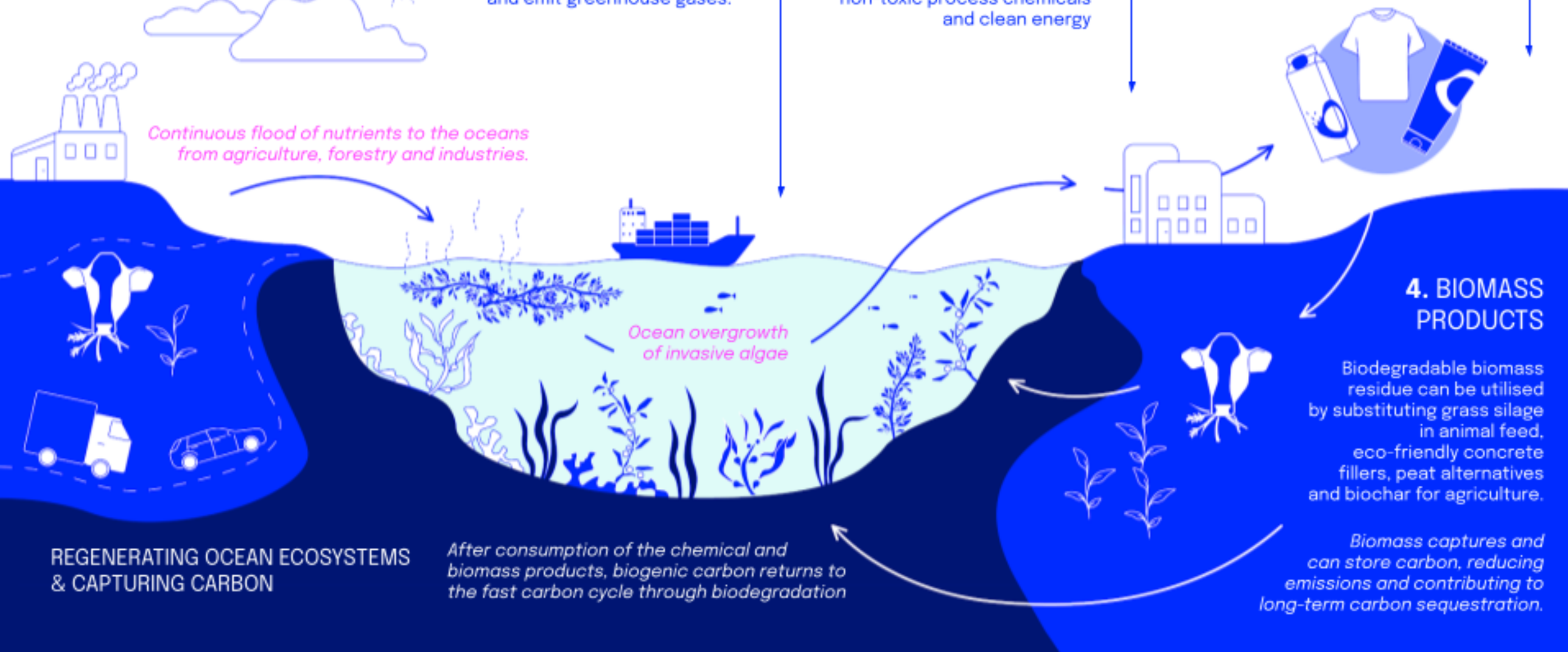
Environmentally sustainable technologies, multiproduct process value and resource utilization maximization

Closed loop production with non-toxic process chemicals and clean energy

3. CHEMICAL PRODUCTS

Biodegradable products enable substituting existing oil-based chemicals in everyday goods.

Decarbonizing consumer products and reducing the carbon footprint.



4. BIOMASS PRODUCTS

Biodegradable biomass residue can be utilised by substituting grass silage in animal feed, eco-friendly concrete fillers, peat alternatives and biochar for agriculture.

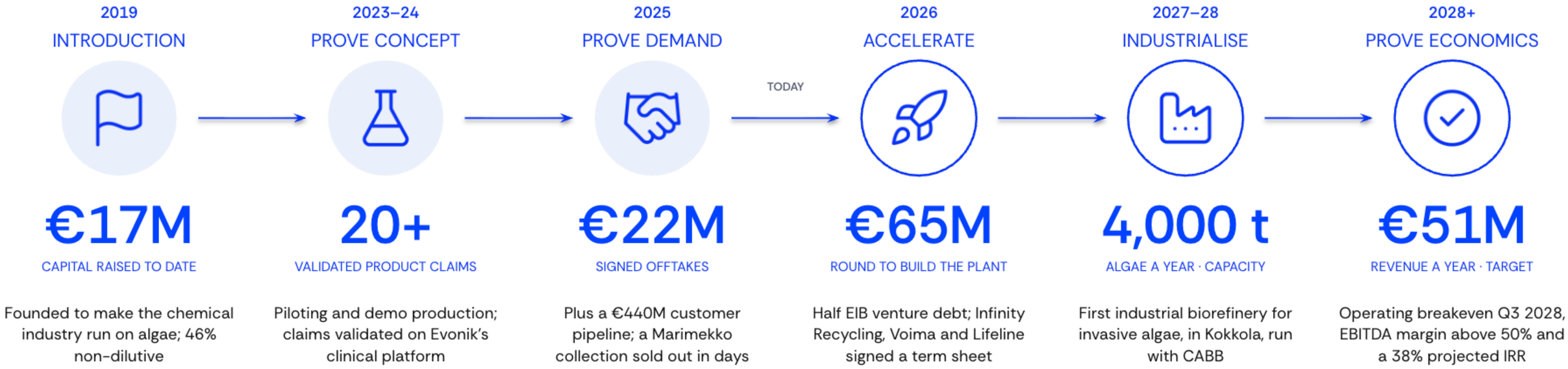
Biomass captures and can store carbon, reducing emissions and contributing to long-term carbon sequestration.

REGENERATING OCEAN ECOSYSTEMS & CAPTURING CARBON

After consumption of the chemical and biomass products, biogenic carbon returns to the fast carbon cycle through biodegradation

From a Biological Insight to an Industrial Business

The next step is to prove that this model can work at industrial scale, combining environmental benefits with customer demand, reliable production and commercially viable economics.



Functional ingredients that can replace existing fossil-based chemicals offer a route into established markets, supported by reported offtakes and a customer pipeline.

Regenerative by Design

Origin by Ocean's model connects environmental restoration with industrial innovation.

By turning invasive algae into valuable ingredients, replacing fossil-based chemicals and finding uses for residual biomass, the company aims to create value at every stage of the process, linking ocean regeneration with a scalable commercial business.

01 — SUPPLY

Turning underutilized waste into a resource

The Sargassum is harvested

02 — PRODUCTION

Biorefinery in Finland (Kokkola)

Closed loop production with non-toxic process chemicals
and clean energy

03 — USE

Replacing oil-based chemicals

Bio-based ingredients in everyday products alginate, in (cosmetics, food, textile and fucoidan in cosmetics anf nutraceutical

04 — RETURN

Biogenic carbon returns to the fast carbon cycle

Biodegradable biomass residue can replace grass silage in animal feed, eco-friendly concrete fillers, peat alternatives and biochar for agriculture.

Rather than treating invasive algae and residual biomass as waste, Origin by Ocean aims to use them as inputs for valuable products and materials.

THE INVESTOR TAKEAWAY

Look for regeneration that can scale



For investors, the key is to look beyond the environmental ambition and understand how the model can scale: from proven technology and customer demand to industrial production, resource efficiency and viable economics.

01 LOOK FOR A MODEL THAT RESTORES, NOT JUST REDUCES HARM

The strongest regenerative businesses address an existing environmental problem while creating a commercially useful product.

02 LOOK FOR A ROUT INTO EXISTING MARKETS

Replacing established ingredients can be more commercially practical than asking customers to adopt entirely new products or systems.

03 LOOK FOR PROOF THAT THE INDUSTRIAL MODEL WORKS

Technical validation and customer interest matter, but the next test is whether production can scale reliably and deliver the projected economics.

WHY IT WORKS

The model links environmental restoration with product performance and resource recovery, creating several potential sources of value from the same biomass.

“Most sustainability stories are a trade-off. You pay more, or you accept less performance. Not this one. The raw material is invasive Sargassum...

Removing it is the whole point. So volume growth and ocean impact pull in the same direction instead of against each other.”

Inka Mero
Voima Ventures



BANKABILITY INDICATORS

1. COMMERCIAL DEMAND ALREADY EMERGING

€22M in reported offtakes and a €440M customer pipeline indicate market interest in bio-based ingredients that can replace fossil-derived chemicals in established markets.

2. CAPITAL TO BUILD INDUSTRIAL SCALE

€17M raised to date and a planned €65M investment support the development of a first-of-its-kind biorefinery in Kokkola, Finland. Successful execution will depend on delivering reliable production at the planned scale.

3. PROVE INDUSTRIAL ECONOMICS

With a target of €51M in annual revenue and operating break-even projected for Q3 2028, bankability depends on validating production costs, achieving planned capacity and delivering the expected margins.

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A Nordic initiative turning circular ideas into investable business

NORDIC CIRCULAR INVESTMENTS